

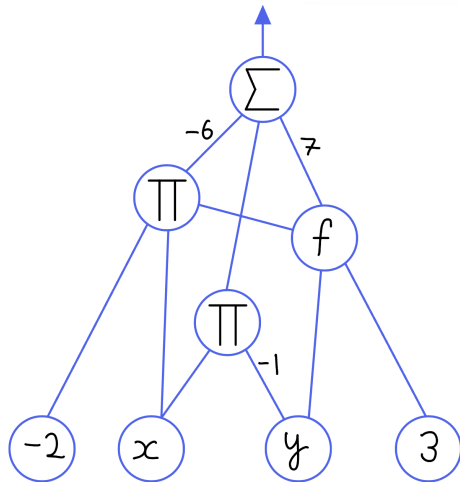
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Debordering Closure Results in Determinantal and Pfaffian Ideals

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Algebraic Circuits with Oracles



We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input-output path

Figure: $-6(-2x \cdot f(y, 3)) - xy + 7f(y, 3)$

Circuit Complexity for Ideals

In algebraic complexity, we are interested in characterizing the *circuit complexity* of some family of polynomials.

Definition

Fix some polynomials $g_1(\bar{x}), \dots, g_k(\bar{x}) \in \mathbb{F}[\bar{x}]$.

The *ideal generated by* $g_1(\bar{x}), \dots, g_k(\bar{x})$ is the set of combinations

$$\langle g_1, \dots, g_k \rangle = \left\{ \sum_{i=1}^k h_i(\bar{x}) \cdot g_i(\bar{x}) \mid h_i(\bar{x}) \in \mathbb{F}[\bar{x}] \right\}.$$

Question

Suppose $f \in \langle g_1, \dots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$?

Principal Ideals

Example

The *principal ideals* are generated by a single polynomial g .
If $f \in \langle g \rangle$, then g is a *factor* of f .

Question

Suppose $f \in \langle g \rangle$. Does g have a small f -oracle circuit?

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If g is a factor of f , $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$.

Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0, g can be *border computed* by a circuit of size $\text{poly}(\text{size}(f), \deg(f))$.

By border computation, we mean the circuit computes the following:

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon) \in \mathbb{F}(\varepsilon)[\overline{x}], \quad q \geq 1.$$

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Closure Results in Determinantal Ideals

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. Then there is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant for some $t = n^{\Theta(1)}$.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *border computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = \Theta(r^{1/3})$.

Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(\overline{x})$ in a standard basis indexed by combinatorial objects, and leverage the combinatorics to talk about specific terms.
- To get a circuit for a specific basis term, we use *Homogenization* as well as the *Isolation Lemma*.

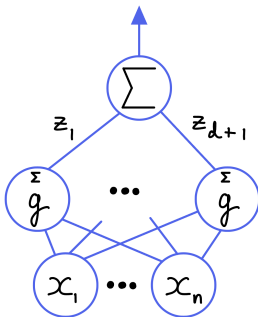
Homogenization

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

Lemma (Folklore)

Say g is computed by a size s , depth Δ f -oracle circuit with top Σ -gate.



Then, we can compute $\text{coeff}_{t^i}(g)$ by a size $O(ds)$, depth Δ f -oracle circuit.

Issues with Homogenization

If a circuit border computes $g(\overline{x})$, then it computes

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon)$$

for some $q \geq 1$.

Idea: Homogenize with respect to ε .

Problem: q can be *arbitrarily large*

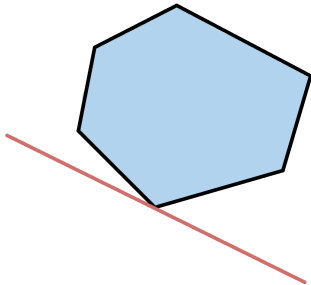
$\implies$ Homogenization gives *large circuit*.

Isolation Lemma

In our proof, we have a specific monomial in $g(\bar{x})$ we want to *isolate*.

Lemma ([KS01, Lemma 4])

Linear programs with *random* cost functions will have a unique minimum.



Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

The Isolation Lemma shows that the $z_1, \dots, z_\ell$ can be choose to be small
 $\implies$ unique lowest $\deg_w$ -term $\implies$ homogenization w.r.t w is small.

Thank You!

Bibliography I



Robert Andrews and Michael A. Forbes.

Ideals, determinants, and straightening: Proving and using lower bounds for polynomial ideals.

In *STOC '22: 54th Annual ACM SIGACT Symposium on Theory of Computing, Rome, Italy, June 20 - 24, 2022*, pages 389–402. ACM, 2022.



Peter Bürgisser.

Completeness and Reduction in Algebraic Complexity Theory.

Springer Berlin Heidelberg, 2000.

Bibliography II



Peter Bürgisser.

The complexity of factors of multivariate polynomials.

Foundations of Computational Mathematics, 4(4):369–396, September 2004.



Anakin Dey and Zeyu Guo.

Debordering closure results in determinantal and pfaffian ideals.

ITCS, 2025.

Bibliography III



Joshua A. Grochow.

Complexity in ideals of polynomials: Questions on algebraic complexity of circuits and proofs.

Bulleting of the EATCS, 131, 2020.



Adam R. Klivans and Daniel Spielman.

Randomness efficient identity testing of multivariate polynomials.

In *Proceedings of the 33rd Annual ACM Symposium on Theory of Computing*, pages 216–223, 2001.